The role of Camera-Based rPPG in Musical Biofeedback Systems for Stress Reduction and Consciousness Regulation

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Abstract—In our modern world, many people experience stress on a daily basis, which can have detrimental effects on both physical and mental health. This paper analyzes a theoretical framework consisting of contactless pulse measurement (rPPG) in combination with musical biofeedback for stress reduction and consciousness regulation.
The research concludes that the CHROM algorithm is best suited for this approach due to its efficiency and robustness. Furthermore, the results indicate that the sonification strategy should be based on the principles of ecological perception, utilizing organic nature sounds that the individual finds pleasant. Similar studies have shown that this combined approach effectively reduces stress and leads to a higher state of conscious awareness. This work provides a theoretical basis for implementing accessible, hardware-independent biofeedback systems in everyday environments.

Index Terms—Remote Photoplethysmography, Computer Vision, Musical Biofeedback, Stress Reduction, Heart Rate Variability, Self-Regulation, Consciousness States

1 Introduction

Stress can have detrimental effects on the human body, ranging from depression, anxiety and social withdrawal to sleep deprivation and obesity. Stress has a direct impact on the cardiovascular system [1]. A wide variety of methods exist to monitor these cardiovascular signals and to track stress levels. The measurement of pulse waves is called plethysmography. Photoplethysmography (PPG) utilizes light reflectance to detect physiological changes. A traditional method for PPG is contact-based sensors, such as pulse oximeters [2]. Remote Photoplethysmography (rPPG) uses standard video cameras to monitor the skin color and filter for subtle skin-color variations that are caused by flowing blood. Based on this information, the heart rate can be calculated [2]. These measurements can be directly reported back to the user using biofeedback. Biofeedback describes a therapeutic concept in which a user receives real-time information about their physiological processes, such as respiration or heart rate. This data is typically made accessible through an output device, such as a visual display. Alternatively, the information can be translated into different forms, such as auditory signals. The primary goal of biofeedback is to make these biological processes perceptible, enabling the user to gain conscious awareness and control over functions that were previously automatic [3].

The objective of this paper is to evaluate a camera-based stress-tracking system combined with musical biofeedback. The goal is to analyze whether such a system can be utilized to lower stress levels and regulate consciousness.

In the following chapters, each individual component of this system is investigated. The second chapter focuses on the physiological reactions of the body when stress is perceived and its corresponding impact on the heart. The third chapter examines how stress can be measured using a remote camera. The fourth chapter goes into detail regarding why simple music listening without biofeedback is insufficient. The fifth chapter explains strategies for converting these measurements into sound. Finally, the sixth and seventh chapters explain how biofeedback leads to regeneration and the regulation of consciousness.

2 Stress and the autonomic nervous system

This chapter starts with a deeper look at stress and its impact on the autonomic nervous system. It explains why stress has an impact on the cardiovascular system. Moreover, it differentiates between the heart rate and the heart rate variability.

The autonomic nervous system consists of the parasympathetic and the sympathetic branches. The parasympathetic system acts during periods of rest and calm and regulates functions such as rest and digest. In contrast, the sympathetic system is activated in situations in which we perceive a threat. In these cases, the heart rate increases to release more energy [4].

McEwen (2007) describes this as the allostatic effect. The body dynamically adjusts its internal state to meet the standards of the environment. For instance, in a life threatening situation a lot of energy is released. The short-term mobilization of energy is a protective reaction of the body. If this activation occurs chronically without enough recovery, it can lead to permenant damage (Figure 1) [1].

Individual differences in stress perception are rooted in the brain's ability to evaluate environmental stimuli. According to McEwen (2007), the brain exhibits significant plasticity, meaning its responses to stressors are shaped by genetics, early development, and past experiences. If an individual has for example experienced a trauma, the brain's system may become hypersensitive [1]. It might

trigger a 'fight or flight' response in harmless situations [4]. This can lead to coping strategies, such as smoking or drinking. These coping strategies try to calm down the sympathetic branch of the autonomic nervous system. On the other hand, the plasticity also allows for an intervention in the other direction. The brain can be trained to shift from a threatening state to a calm state [1].

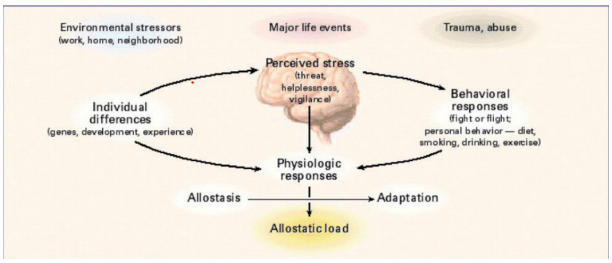

Fig. 1. The figure illustrates the mechanism of allostatic load and the underlying pathways of stress perception and physiological adaptation [1].

When monitoring the heart, it is crucial to differentiate between the heart rate and the heart rate variability. The heart rate can be thought of as the quantity and the heart rate variability as the quality of heart activity [5].

The heart rate measures the total number of heartbeats per minute. It reflects how fast the heart is beating at any given moment. However, it can be described as a relatively coarse metric. Therefore, it is a releative insensitive marker for early-stage stress [5].

Heart Rate Variability (HRV), on the other hand, focuses on the smaller level of cardiac timing. It describes the fluctuation in the time intervals between two heartbeats. This is defined as Interbeat Intervals (IBIs). HRV consists of the constant changes in these time intervals from one beat to the next [5].

From a physiological standpoint, a healthy biological system needs to be highly adaptable. This means that a healthy heart does not maintain a perfectly rigid or identical time between intervals. Instead, the autonomic nervous system constantly changes the timing of each beat to respond to internal and external demands [5].

If the intervals between heartbeats become too consistent and lose this natural irregularity, it is often a clinical sign of physiological strain or a lack of regulatory flexibility. High variability indicates that the body is in a state of balance and can rapidly adjust to stressors. Therefore, while HR measures the quantity of beats, HRV reflects the quality and efficiency of the autonomic nervous system's control over the cardiovascular system [5].

3 Remote Photoplethysmography

To measure the heart beat and the heart beat variability a technology called remote Photoplethysmography can be employed [2].

First, the theoretical foundations are explained, followed by a discussion on the optimal regions of interest within the facial area. Based on these regions, various approaches for detecting and tracking the ROI are presented. Furthermore, the chapter examines traditional methods of feature extraction before discussing modern artificial intelligence-based paradigms.

3.1 Theoretical Foundations

The foundational work for modern remote photoplethysmography (rPPG) was made by Verkruysse et al. (2008), who demonstrated that the heart beat could be monitored using a standard digital camera. This relies on the fact that hemoglobin absords light differently than the neighboring body tissue. As the blood volume changes in the body with every heartbeat, the intensity of the reflected light varies. Their research identified that the green color channel contains the strongest information regarding the heart beat (Figure 2). Despite the successful proof-of-concept, two challenges were highlighted: Motion artifcats caused by movement and noise due to the camera's sensor [2].

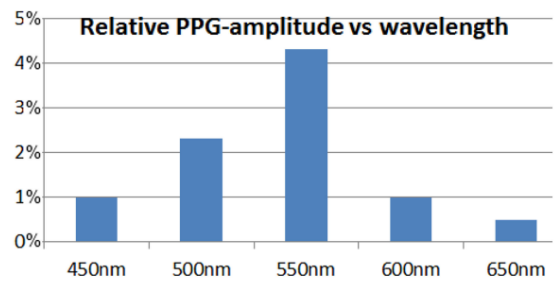

Fig. 2. Spectral sensitivity of the PPG signal. Green light exhibits the highest amplitude, making it the most reliable channel for non-contact heart rate monitoring. [6]

3.2 Face Detection and region of interest selection

The selection of the right region of interest for rPPG measurements is critical for the suppression of motion and illumination artifacts. Previous approaches often utilized the entire face as a global ROI for signal extraction. However, Lempe et al. (2013) demonstrated that the signal quality varies substantially across different facial areas [7].

To quantify this, they implemented a grid layout over the face and calculated the local signal-to-noise ratio for each segment. Through this spatial analysis, the cheek regions were identified as the most promising areas for pulse extraction (Figure 3) [7].

A significant practical advantage of the cheeks is that they are rarely covered by clothing or facial hair, which supports a more stable measurement compared to a global face ROI. These findings highlight a significant technical challenge. The requirement for advanced segmentation and tracking algorithms that can reliably detect and follow the cheek area, even during head movements, to ensure a consistent and high-quality pulse signal [7].

3.2.1 Landmark Tracking

Lempe et al. (2013) identified the optimal regions for extraction. The challenge is to keep these region of interests stable and to handle noise due to motion and light changes. Especially in the use case of human-computer

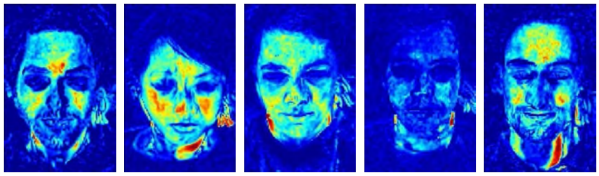

Fig. 3. The quality of the rPPG signal in the face. Blue shows low activity and red hight activity [7]

interaction, brightness changes lead to severe fluctuations of pixel values in the face, examples include watching a movie or gaming. Li et al. (2014) propose a framework to solve these issues. First of all, the viola-jones detector is used to get a rough estimation of the face location. Then, 66 facial landmarks are placed, and based on these landmarks, a precise mask for the region of interest is generated.To maintain a stable measurement, the Kanade-Lucas-Tomasi (KLT) feature tracker is employed. KLT points are used to continuously track the face, ensuring it is also tracked when moved. A second important step is the correction of brightness changes. It is assumed that these changes are not only visible in the face but also in the background. The average brightness changes in both the background and the face are measured, and a normalized least mean squares filter is used to find the optimal coefficient to subtract background noise from the facial area. Furthermore, non-rigid movement like smiling or speaking induces changes in the skin which cannot be compensated through tracking alone. Consequently, the measurements are split into small segments, and those with a high standard deviation are discarded. This framework was tested with the sophisticated MAHNOB-HCI database and reached a low error score of 6.87% with 527 users. Despite these achievements, there are still quite a few problems; for instance, large head movements lead to a significant loss of KLT tracker points, which results in an unstable localization of the ROI [8].

3.2.2 High-Density 3d Face Mesh

A limitation of traditional landmark-based tracking systems is that they have a reduced robustness against head rotations or non-rigid deformations. To overcome these hurdles, there have been approaches to use artificial intelligence to reach a higher resolution. The previous model by Li et al. used 66 vertices, while this new approach by Kartynnik et al. (2019) uses a dense 3D mesh consisting of 468 vertices. This model creates a 3D mesh of the face, meaning it can estimate the depth of each point even from a normal 2D camera (Figure 4). Depending on the hardware, it can deliver 100 to 1000+ FPS [9].

Different models that also use many vertices often use standard methods based on averaging effects. They don't work well when there are small inaccuracies, like the closing of one eye. Due to the architecture of the neural network developed by Kartynnik and his team, it is possible to use it in real-time. It has been trained on a dataset of around 30,000 images. It has been tested that it has the same accuracy as human experts that model the face [9].

Fig. 4. Example of a face mesh calculating with an artificial intelligence model [9]

3.3 Traditional Frameworks for signal extraction

Traditional rPPG frameworks typically follow a multi-stage process. First, the face is detected and tracked, often using the Viola-Jones algorithm or similar cascade classifiers to define a region of interest. Then methods such as Principal component Analysis (PCA) or Independet Component Analysis (ICA) are employed to isolate the pulse signal from the green color channel. While effective if the person remains still, these approaches struggle with motion artifacts, as they require quite a few seconds of stable data to compute the signal components [10].

3.3.1 Blind Source Separation

Poh et al. (2010) introduced blind source separation (BSS) to rPPG. Their framework assumes that the recorded RGB channels do not only contain the pulse information but also significant noise in the form of light changes and movement. The challenge is that this noise has to be separated from the real physiological information. Mathematically speaking, the resulting color channels are treated as a linear mixture [11]:

$$x(t) = As(t)$$

The goal is to find a demixing matrix W, so that the original source signals $\hat{s}(t) = Wx(t)$ can be calculated. To find W, it is important to search for statistical independence between the sources. For that, the JADE algorithm (Joint Approximate Diagonalization of Eigenmatrices) is used. Unlike deterministic methods, JADE is an iterative process that tries out different combinations to maximize non-Gaussianity, which results in a higher computational load and increased latency (Figure 5) [11].

For the initial localization, the viola-Jones algorithm is used to detect the face. In this specific setup, only 60% of the bounding box width is used as the Region of Interest. The raw signals from this ROI are then spatially averaged and normalized using Zero-Mean and Unit-Variance to balance different channel intensities [11].

After the independent components are recovered, the pulse is extracted by using the fast fourier transform. An operational range from [0.75 - 4.0] Hz is allowed, because this corresponds to 45 - 240 BPM, the range a human pulse usually lies in. Additionally, a temporal consistency threshold is implemented.A jump of only 12 BPM is allowed between consecutive measurements that are taken

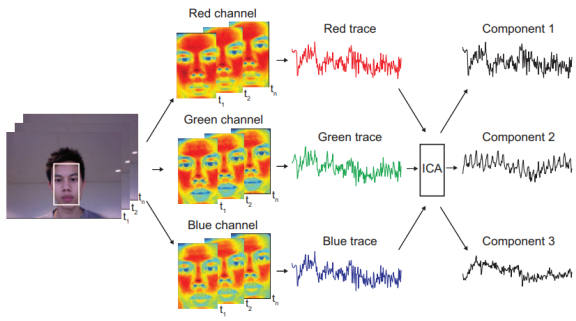

Fig. 5. Seperation of color channels [11]

one second apart. Anything different is rejected as noise or an artifact. The highest power peak within the frequency spectrum is chosen as the final heart rate [11].

To compare the accuracy of this methodology to a gold standard Bland-Altman plots are used. These plots show that ICA can significantly reduce the Root Mean Squared Error (RMSE) from 19 BPM to about 4.6 BPM even when the subject is moving [11].

However, there are physical limitations to consider. According to the Beer-Lambert Law, the way light travels through and is absorbed by blood vessels is technically nonlinear. But for the sake of computation, ICA uses a linear approximation, which works well for short time windows. The system is also highly dependent on the ambient lumination. If it is too dark, the SNR drops too low for a reliable measurement [11].

Another risk is that the wrong component is picked. In this study, the second independent component is usually chosen as the pulse signal because hemoglobin absorptivity is highest in the green spectrum. However, this fixed selection could be wrong if the noise is too strong. Finally, it is noted that using simple webcams is much more cost-effective than using expensive thermal cameras, and the software-based approach even allows for the simultaneous detection of multiple people in one frame [11].

3.4 The Chrominance-based Method

Blind source separation (BSS) techniques provide a powerful mathematical tool for signal isolation. Nevertheless, these techniques face significant challenges in scenarios with substantial motion and light artifacts. To address these limitations, de Haan and Jeanne (2013) proposed a physics-based alternative known as the Chrominance-based method (CHROM). It effectively addresses the challenge of subject motion when calculating the heart rate, as periodic movements can render simple intensity-based selections useless. Furthermore, while many algorithms require long observation intervals to achieve sufficient frequency resolution, CHROM is designed to be highly efficient. Similarly to traditional methods like PCA or ICA, the CHROM algorithm utilizes the fact that blood absorbs a large percentage of green light. Additionally, it analyzes how light is reflected by human skin. According to the dichromatic reflection model, this reflection consists of a specular and a diffuse component (Figure 6).According to the model of de Haan, a measured signal C_i for every color channel can be written as [6]:

$$C_i = I \cdot (\rho C_{dc} + \rho C_i + s_i)$$

Here, ρC_{dc} represents the constant color of the skin, ρC_i the pulsating change due to blood flow and s_i the specular reflection. Only the diffuse reflection enters the skin and carries the color of the blood. The specular reflection, however, does not enter the skin. It is reflected directly off the surface and thus contains no information about the blood. Since this specular part is an additive value, a simple division of color channels cannot cancel this effect. Therefore, CHROM uses a subtractive approach [6]:

$$(R + noise) - (G + noise) = R - G$$

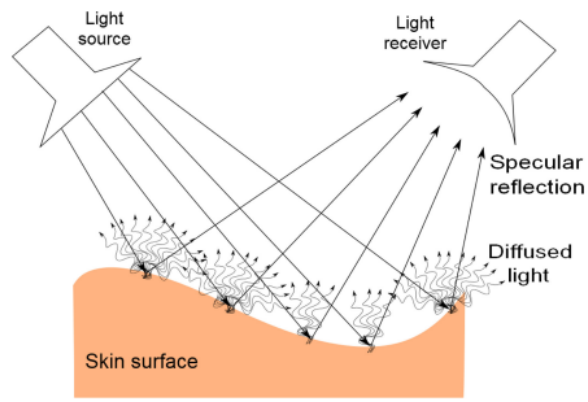

Fig. 6. The illustration depicts light reflection on the skin [6]

Light intensity changes (I) and specular reflections (s_i) influence all color channels proportionally. Consequently, these noise vectors primarily move along a single axis in the color space. To reduce these effects, CHROM projects the normalized RGB signals onto a plane that runs perpendicular to the standard skin color vector. Based on empirical data from a diverse set of subjects, de Haan and Jeanne derived a standardized average vector for skin color: $[0.7682, 0.5121, 0.3841]$. By projecting the signals onto the orthogonal X and Y planes using this vector, the diffuse and specular light components cancel each other out. The resulting signals X_s and Y_s are thus largely free of luminance changes [6]:

$$X_s = 3R_n - 2G_n$$

$$Y_s = 1.5R_n + G_n - 1.5B_n$$

These two orthogonal planes can be comined with a factor α. This represents the light absorption of hemoglobin [6]:

$$S = X_s - \alpha \cdot Y_s \quad where \quad \alpha = \frac{\sigma(X_s)}{\sigma(Y_s)}$$

Even after projection, the signal S may contain a slow drift caused by respiration or minor posture changes. To isolate the cardiac signal, a Bandpass Filter is applied, typically tuned to a range of 0.7 Hz to 4.0 Hz (42 to 240 BPM). Finally, a Fast Fourier Transform (FFT) is performed on the filtered signal to identify the dominant peak in the

power spectrum. For instance, a peak at 1.2 Hz corresponds to [6]:

$$1.2Hz \times 60seconds = 72BPM$$

One of the most significant advantages of CHROM is its temporal efficiency. While traditional BSS methods (Blind Source Separation) like ICA or PCA often require up to 30 seconds of data to stabilize, CHROM utilizes an Overlap-Add (WOLA) procedure. By using overlapping windows of only 1.6 seconds, the system provides a continuous, near-instantaneous pulse estimate [6].

In comparative studies, the robustness of CHROM is evident. CHROM has a high stationary accuracy and predictes the correct pulse with a 98% accuracy. It has a 92% correlation with the medical-grade contact sensor. Furthermore, it is resilient against motion. Under heavy motion, e.g., on a fitness stepper, it maintains a 48% success rate. ICA and PCA fail significantly, and only reach 4% and 11%. Moreover, CHROM can also process different skin tones. Darker skin tones naturally lower the Signal-to-Noise ratio because they absord more light. Chrom remains the most robust classical algorithm across all phenotypes [6].

3.4.1 Modern Deep Learning Paradigms

While physics-based methods like CHROM improved the handling of light and motion artificats, they are still based on fixed mathematical assumptions. To overcome these hurdles the field has increasingly shifted towards artificial intellligence models [10].

Following the initial proof-of-concept, a significant increase in rPPG research was observed, particularly after 2017. This trend is closely linked to the rapid improvements in artificial intelligence and deep learning. In 2025 82.05% [10] of papers about rPPG technology used an approach with artificial intelligence. A major problem is the availability of diverse, large-scale datasets. The available sets only have data of people from North America, Europe and China. This leads to a huge underrepresentation of various age groups and diverse skin pigmentations [10].

With the rapid advancement in artificial intelligence, modern rPPG systems have shifted towards end-to-end models. These architectures can process raw data or image sequences to estimate heart rate directly, often without the need for manual face isolation or region of intrest selection. There are three architectural paradigms: Convolutional Neural Networks (CNNs), Hybrid Neural Networks and Transformed-based architectures [10].

In modern rPPG systems, convolutional neural networks (CNNs) are the standard for processing image data. Their main task is to recognize the very subtle color changes in the skin pixels that happen during a cardiac cycle [10].

A big problem in this field, is that there aren't enough large and diverse datasets to train these models from scratch. Because of this, researchers often use transfer learning [10]. The idea here is to take a model that was already trained on a different task, such as general face recognition. By freezing the first layers and only fine-tuning the last ones with rPPG-specific data, the model can use its existing knowledge of facial features. This approach makes it possible to get good results even with smaller datasets and saves a lot of training time [12]. Furthermore, researchers are increasingly focused on developing leaner architectures that can be deployed on mobile devices [10].

In addition, a different category of neural networks has emerged: Hybrid Neural Networks. These combine different neural networks like CNNs, RNNs and autoencoders to form a more robust system [10].

In the past years powerful large language models like ChatGPT have emerged that use transformers to understand words and sentences. The same technique can be employed to analyze pictures. The picture is broken up into small rectangular patches, which the network analyzes in relation to one another [13]. By Comparing different regions simultaneously, the transformer is able to differentiate between the heart beat and noise, such as micro-movements. This leads to a more robust heart rate estimation [10].

3.4.2 Summary of rPPG Frameworks

There are lot of different approaches to calculate the heartbeat with an rPPG technique. Modern deep learning algorithms have emerged in the recent years They show promising results, although there are not enough diverse datasets [10]. On the other hand, the CHROM algorithm is promising for a realtime evaluation of data. Traditionally algorithms like PCA are slow and not so precise [6]. Although, there has been a huge advancement in the past years, there are still no long-term studies that monitor the heart beat over several hours. Most researches only test rPPG for a short time, which does not show how it works in real daily life [10].

4 From passive listening to biodfeedback

In the last section it was established how heart rate data is extracted using rPPG techniques. The next step is to analyse how this data can be translated into a piece of music. It is first necessary to review the existing research on passive music listening and its limitation in stress reduction.

According to a meta-analysis by Pelletier (2004), music consistently assists in lowering stress levels. The magnitude of this response is affected by several factors. Patients with prior musical experience, such as playing an instrument, tend to lower their stress levels a lot more, as they often possess a greater predisposition and comfort level with music [14]. Furthermore, Pelletier notes that individual therapy generally results in better relaxation than group session. Although, group sessions are used much more often because they are less expensive. A huge obstacle is that the devices that are used for data collection often cause discomfort. The study highlights that physiological measures or self-reports can cause a disruption of the relaxed state. Behavioral observation seems to collect more accurate data of the the relaxation process [14]. Moreover, the study came to the conclusion that music based on scientific research yiels greater effects for stress reduction than music that was chosen on personal preference [14]. This is likely because preferred music can be too distracting or stimulating. On the other hand, research-based music combined with techniques like verbal suggestion or progressive relaxation achieves the highest impact on decreasing arousal. [14]. In another study, Linnemann et al. examined how music affects stress in daily

life rather than in a laboratory setting. They concluded that while music can effectively reduce stress levels, the magnitude of this effect depends heavily on the listener's intent. Specifically, the largest reduction is achieved when subjects listen with the explicit goal of decreasing stress. This suggests that the underlying motivation is more important for the actual relaxation process than the music itself. Conversely, the study confirms that using music for the purpose of distraction can actually lead to an increase in stress levels [15].

The study of Bernardi et al. (2006) confirms that the results do not depend on the music preferation of the user rather on the BPM and rhythm structure of the music. Additionally, the study found no evidence of habituation, meaning the users do not simply 'get used' to the music over time. Their findings suggest that because the body snychronizes with the music's tempo and rhythm measurable stress-reducing results can occur during passive listening, regardless of the users' conscious intention [16].

5 SONIFICATION

To overcome the limitations of passive music listening, the data must be transformed into an auditory format. This process relies on deliberate mapping strategies to ensure the physiological informations remains interpretable. According to Dubus and Bresin (2013), the effectiveness of such a system depends on how physical quantities are mapped onto acoustic dimensions like pitch, loudness, or timbre to eable accurate data perception [17].

Dubus et al. (2013) performed a systematic review on sonification. They mention that sonification can enhance the perception of one's own body and is therefore widely used in rehabilitation and training. Among different sonification techniques, parameter mapping is especially promising because it allows for continuous transmission of information. In this process, a certain set of mappings between "physical dimensions" and "auditory dimensions" is defined [17].

To establish these mappings, a three-stage process is required. First, the specific auditory dimensions to be used must be identified, such as pitch or timbre. Then the polarity has to be defined. This determines the direction in which the auditory dimension changes according to the data. For instance, if the heart rate increases, it must be decided whether the pitch should increase or decrease. Finally, the psychophysical scaling has to be addressed. Since humans perceive changes in sound non-linearly, it is necessary to define the magnitude of the auditory change relative to the data change [17] [18].

According to Dubus and Bresin (2013), many sonification approaches implement a strategy called "ecological perception". This means that certain sonification principles are part of our subconscious mind. This applies, for example, to loudness, which can be explained by the inverse distance law. A decrease in sound intensity is naturally associated with an increasing distance from a source. Such natural associations are deeply ingrained in human perception and should be prioritized when employing a sonification approach [17].

To ensure a clear understanding of the mapping process, it is essential to define these auditory dimensions. Timbre refers to the perceived quality or 'color' of a sound, which is often used for the dimension 'Time'. In contrast, loudness relates to the sound's intensity or volume, making it the preferred mapping for 'Kinetics' to represent energy or force. Furthermore, pitch that is revered to as the perceived frequency of a sound, is commonly used to represent vertical data, while spatialization refers to the perceived location of a sound source in a 3D environment, often used to depict horizontal data [17].

Furthermore, it is pointed out that the sonic material is decisive. If the sounds make the user uncomfortable, this can have drastic consequences for the effectiveness of the system. Therefore, the use of real-world sound samples is often preferred over simple synthetic tones [17].

In addition, it is important to investigate the chosen sonification strategy scientifically. The study highlights that many approaches are currently chosen in an ad hoc or arbitrary manner without verifying their effectiveness through objective tests [17].

6 BIOFEEDBACK-DRIVEN REGENERATION

As explained previously the heart rate is converted to a piece of music. It is now crucial to understand how the music can help the human body to regenerate. According to Lehrer and Gevirtz (2014), the efficiency of biofeedback relies on the synchronization of the cardiovascular and respiration systems [19].

The human cardiovascular system has a certain resonance frequency at which point it exhibits the best effects on the human body. If this state is reached, has maximum internal order and physiological efficiency. This frequency is reached at approximately 0.1 Hz. This is equivalent to roughly six breaths per minute. By breathing at the resonance frequency, heart rate and blood pressure enter a harmonic state. Furthermore, gas exchange in the lungs are also optimised at this state [19].

In addition, the activity of the vagus nerve is stimulated. The vagus nerve is responsible for the communication between the heart and the central nervous system. This makes it possible for the body to lower the heart rate more effectively [19].

Moreover, biofeedback creates a meditative effect that changes the user's state of consciousness. The user has to focus his thoughts and his energy on the biofeedback signal. Thus, it the brain cannot focus on anxious thoughts. This shift in mental focus leads to a state of heightened awareness, which is explored in the following section on consciousness and mindfulness [19].

7 CONSCIOUSNESS AND STATE REGULATION

The meditative effect of focusing on one's heart rate or respiration can be used to enhance consciousness. This process is frequently described through the concept of mindfulness. This makes it possible for individuals to develop a sensitivity to their thoughts and emotional states. Bishop et al. (2004) established this definition. A fundamental requirement is described as the 'non-judgemental-approach". This means that internal states are perceived with curiosity and openness. Internal thoughts and feeling should be accepted as they are [20].

8 CASE STUDIES

In this chapter two case studies will be presented that highlight the effects of using biofeedback. The use different approaches. One uses the heart beat and the other uses EEG. They are both important for validating stress reduction and consciousness regulation.

8.1 The "Unwind" Musical Biofeedback System

The "Unwind" system, developed by Yu et al. (2018), employs a special approach to mapping the heart rate to music. It uses nature soundscapes combined with sedative music. Sedative music is used in music therapy to effectively decrease heart rate and blood pressure. It has a low tempo and a repetitive structure [21].

The system uses the heart rate variability to provide feedback on the user's breathing patterns. The heart rate accelerates during inhalation and decelerates during exhalation. The framework maps these fluctuations directly to a wind soundscape. The wind volume is reduced during inhalation and during exhalation the wind becomes louder. The user should adapt his breathing pattern in a way that a the wind sound is like a smooth, consistent sinve wave. This promotes consistent respiration [21].

The wind provides immediate feedback on the breathing pattern. The overall relaxation is mapped with a special form of heart rate variability. The system calculates the variability with a sliding window of 16 heartbeats. The organic soundscape is based on the intensity of the heart rate variabilitys. If the variability is low, the environment becomes busy and crowded with bird calls and noises. On the other hand, if the variability increases, the soundscape becomes simpler and more transparent. The frequency of animal sounds decreases and the overall volume drops. This rewards the person with acoustic clarity [21].

The system was validated with 40 young adults. A significant decrease in heart rate was found in the music assisted conditions, and the reduction in respiration rate was significantly greater with biofeedback than without. Skin conductance responses, which indicate sympathetic nervous system activity, decreased markedly during the relaxation sessions. In follow-up interviews, 70% of participants reported that the nature sounds allowed them to quickly visualize calming natural scenes, aiding their focus and mood [21].

8.2 Mindfulness

This study investigated if a system with biofeedback could lower the entry barriers of classical meditation. The focus was on improvements in emotional regulation and cognitive performance. The experiment took place over four weeks and involved 40 participants who suffered from mild to moderate stress. The participants were separated into an experimental group and an active control group. As a tracking device, this experiment utilized the Muse™ headband (Figure 7). This is a mobile EEG system connected to a smartphone app. The participants had to do daily, active attention training sessions. The particiapnts were able to gather real-time feedback with the Muse™ headband. The recorded brainwaves were translated into acoustic feedback. The active control group performed identical relaxation exercises for the same duration but without the wearable device or acoustic feedback [22].

Fig. 7. Subject wearing the Muse™ EEG headband during a neurofeedback session. [22]

The results after four weeks were significant and underscore the potential of wearables in professional stress management. The control group did not show large changes. On the other hand, the experimental group recorded a decrease in perceived stress of 7% after two weeks. After weeks it went down to 16%. In addition, participants demonstrated higher concentration levels and better impulse control during complex tasks. Moreover, the participants' brainwaves showed a lasting shift toward a relaxed mind. The central finding of Balconi et al. is that real-time feedback is the decisive factor for success [22].

9 ARGUMENTATION

In this chapter, the final argumentation for the closed-loop system is presented. It discusses whether a camera-based approach using rPPG combined with musical biofeedback is effective for stress reduction and the regulation of consciousness.

First and foremost, stress in this context is defined as the activation of the sympathetic nervous system and the corresponding withdrawal of the parasympathetic branch. These physiological shifts directly affect heart activity. While physical activity is best tracked via heart rate, situations in which the user is sitting still or not moving significantly require the analysis of Heart Rate Variability to accurately quantify stress. Consequently, a comprehensive system must be capable of addressing both states [1] [5].

Both aspects can be effectively monitored using an rPPG approach. For real-time feedback, it is crucial that the underlying algorithms are computationally efficient. Furthermore, the system must remain robust against motion artifacts. In daily life, users do not remain perfectly still for extended periods. Therefore, the CHROM algorithm is ideally suited for this use case. It is fast to calculate as it leverages the standardized optical properties of human skin, and it has been proven to be resilient against motion changes [6]. In contrast, other methods such as blind source separation techniques often require longer observation windows for an initial measurement or demand significant hardware

resources [11] . As highlighted by Debnath (2025), modern deep learning architectures are frequently very resource-intensive, requiring researches to develop specialized models for 'resource-constrained environments' [10]. Compared to these complex neural networks, the CHROM algorithm is significantly simpler to implement and requires far less computational power [6]. This efficiency ensures that the heart rate can be estimated immediately on standard hardware, which is essential for providing synchronized results without processing delays. Additionally, these networks are often trained on datasets that lack a sufficient variety of skin tones. This leads to the conclusion that a CHROM-based approach is best suited for this application [10].

In the next step, this physiological data must be converted into musical biofeedback. The music must be enjoyable to the user to prevent listening fatigue while also avoiding boredom. Passive listening alone is generally not successful in lowering stress. The user's intention is far more critical. The user must actively intend to lower their stress levels and gain better conscious awareness [15]. A continuous mapping process is necessary to provide the user with immediate feedback regarding their own body [17].

To achieve this, the system follows a sonification strategy based on parameter mapping to ensure a conscious transmission of information [17]. In this process, specific auditory dimensions have to be chosen carefully. According to Dubus and Bresin (2013), this should follow the principle of 'ecological perception'. When the stress or the heart beat is low (calm), the sound has to reflect this behavior. This shows that the choice of sonic material is decisive. It is important to verify the chosen strategy scientifically [17].

The "Unwind" system is a practical example of these principles, as it successfully utilized nature sounds, such as wind noises, to imitate breathing patterns. Furthermore, this organic soundscape can be used to reflect the participant's arousal level [21]. This follows the principle of "ecological perception" as mentioned before: if the body is calm, the soundscape reflects this tranquility [17].

Additionally, case studies involving the "Muse" headband have demonstrated that biofeedback can effectively increase conscious awareness and mitigate anxious thoughts. Biofeedback serves as an "anchor," preventing the user from ruminating on distracting or stressful thoughts [22]. This process also enables various health-promoting mechanisms through biofeedback-driven regeneration. The respiratory and cardiovascular systems are synchronized, and when this resonance frequency is reached, the body functions at its optimum. This stimulates the vagus nerve, making it possible to further lower stress levels and promote long-term recovery [19].

10 Conclusion

This paper investigated the theoretical and technical requirements for a biofeedback system designed for stress reduction and consciousness regulation.Based on the research, the CHROM algorithm proved to be the most suitable solution for heart rate detection. It is efficient to implement, requires minimal hardware resources, and provides robust results even when dealing with motion and lighting artifacts. The transformation of the data to sound is most effective when the principles of ecological perception are taken into account. An organic soundscape is necessary to ensure an intuitive and pleasant user experience. In summary, while this work is based on a theoretical analysis of existing methods, it provides a clear roadmap for future implementations of camera-based closed-loop systems.

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